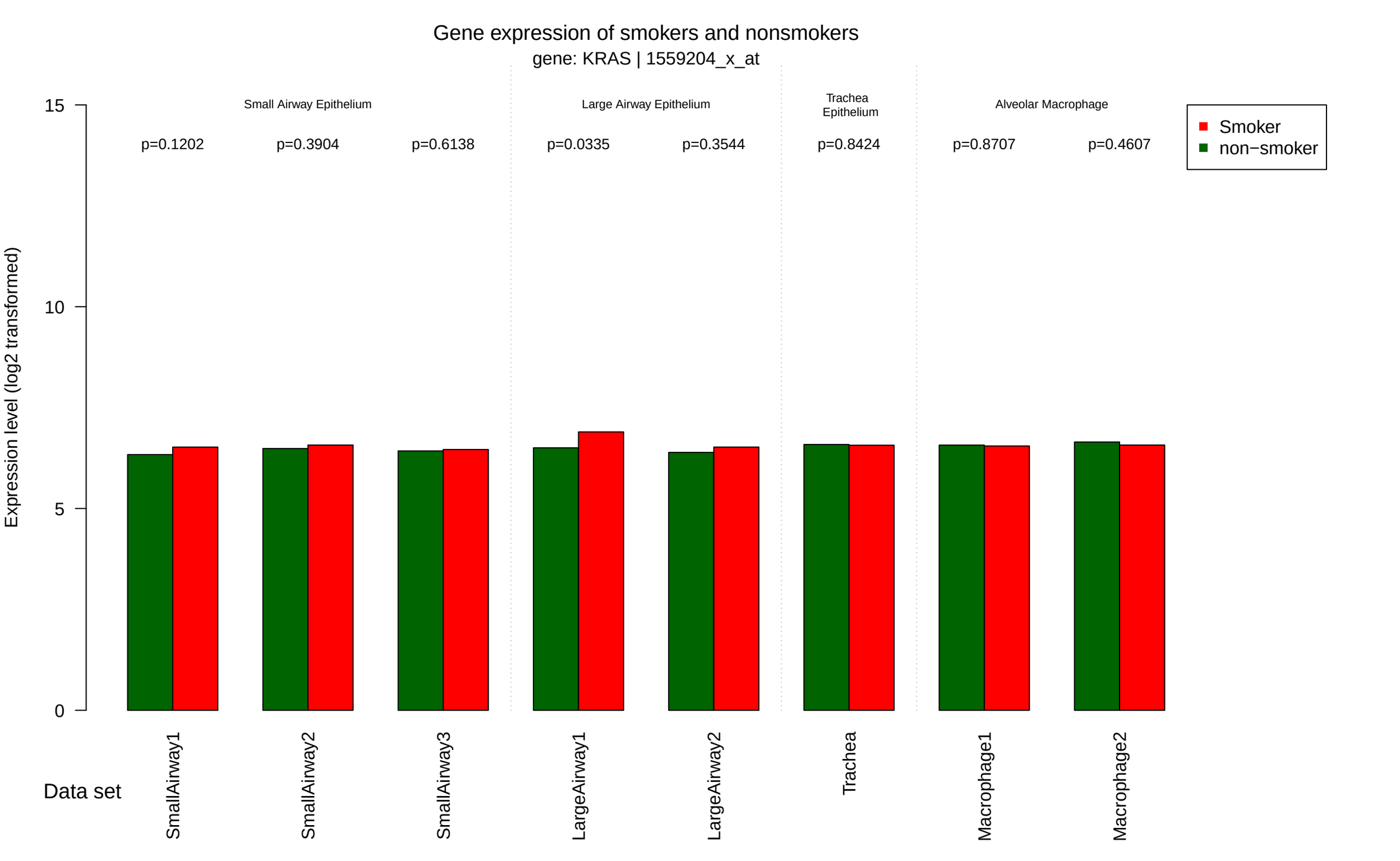
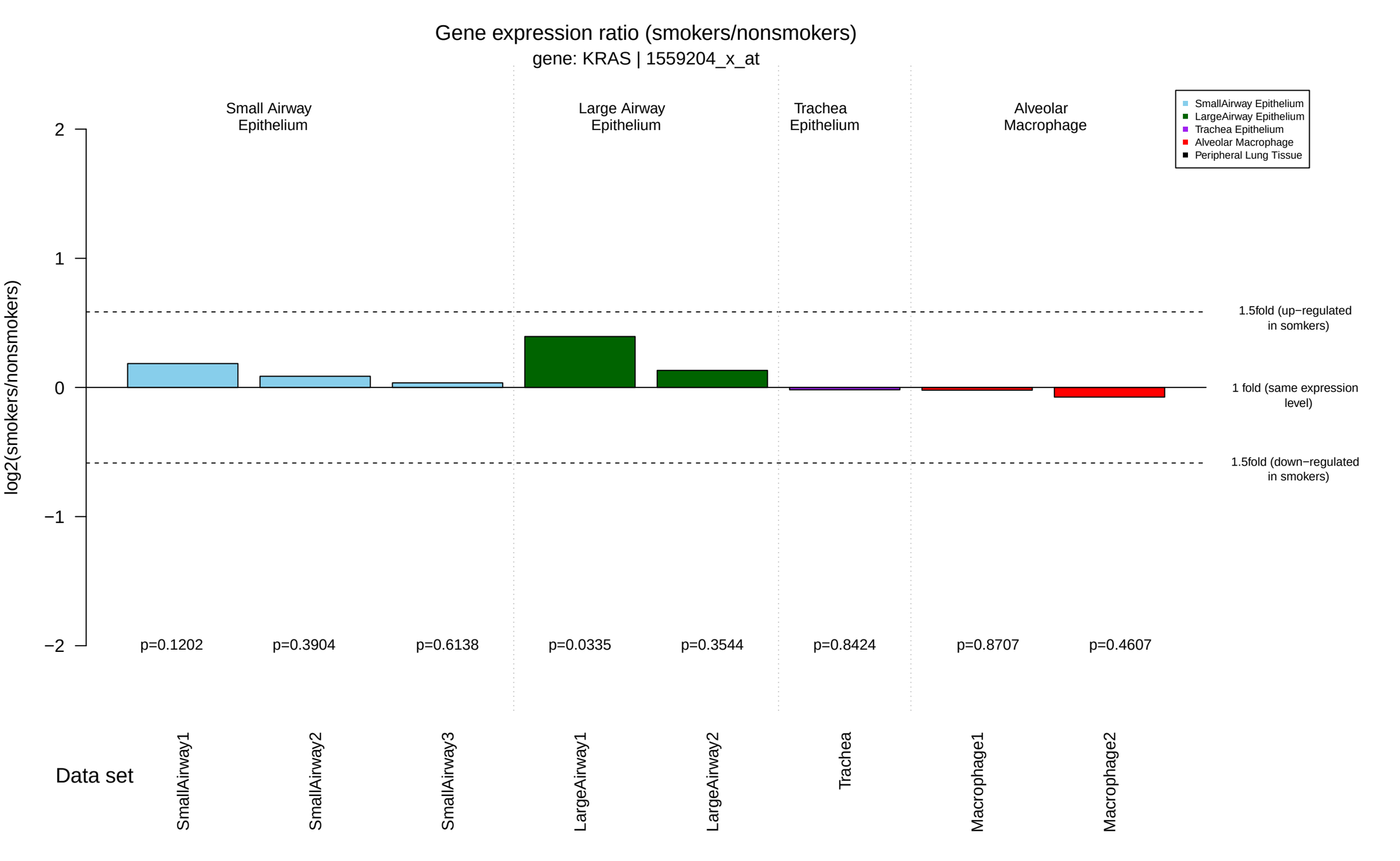
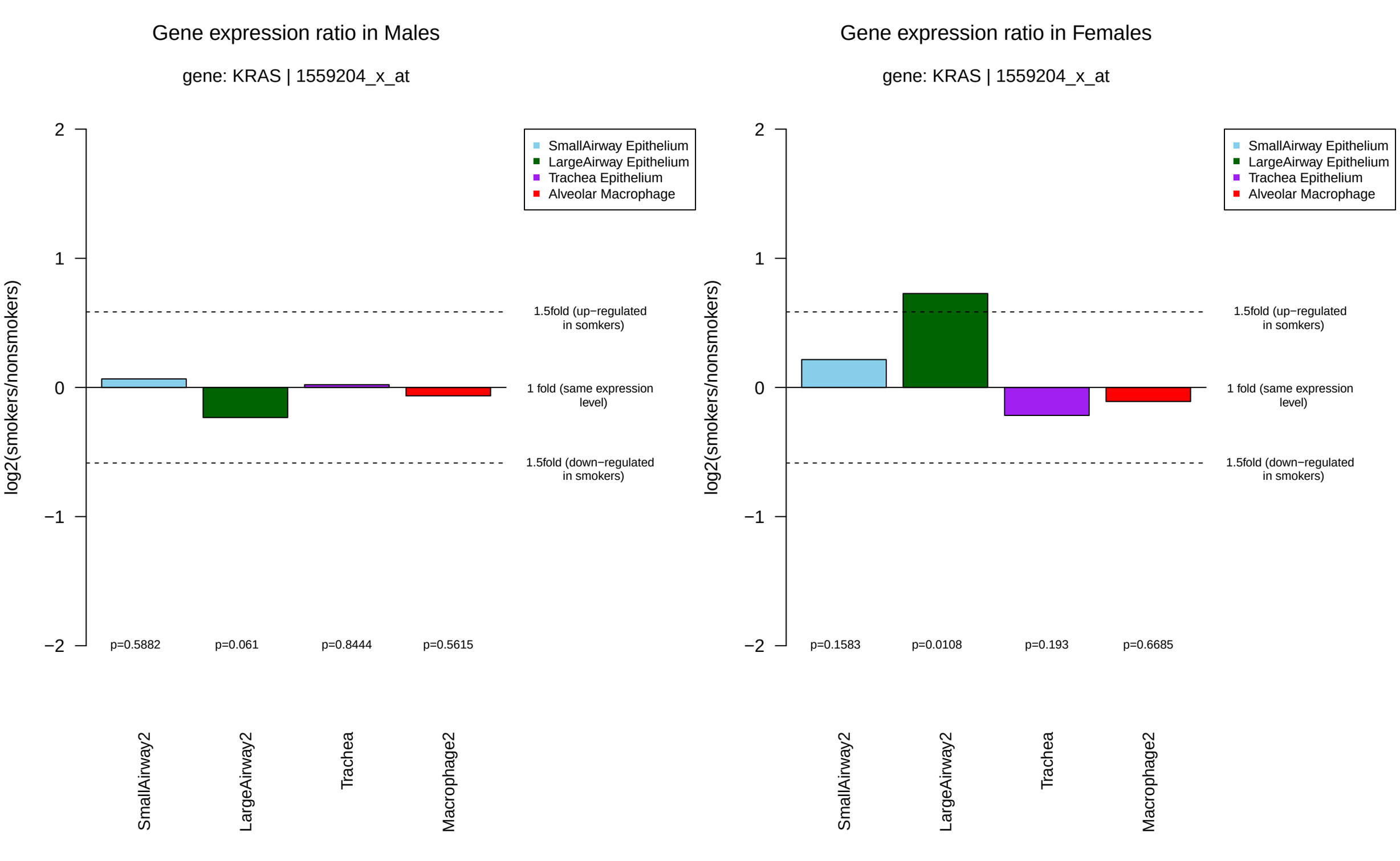
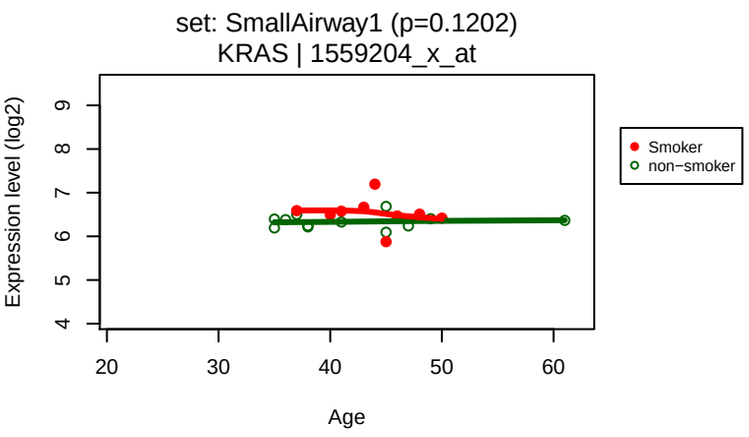
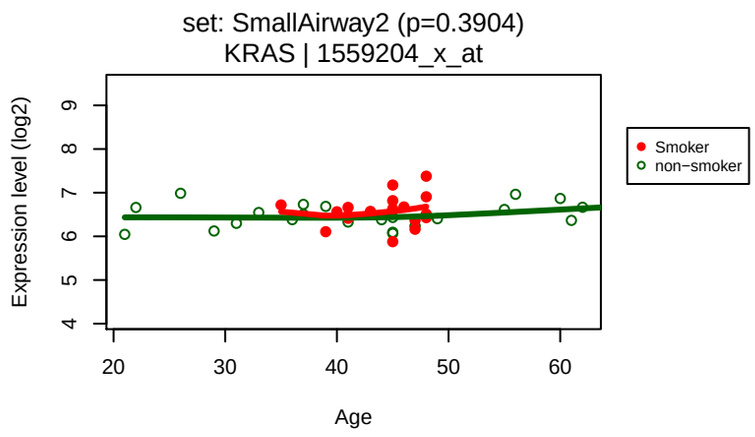
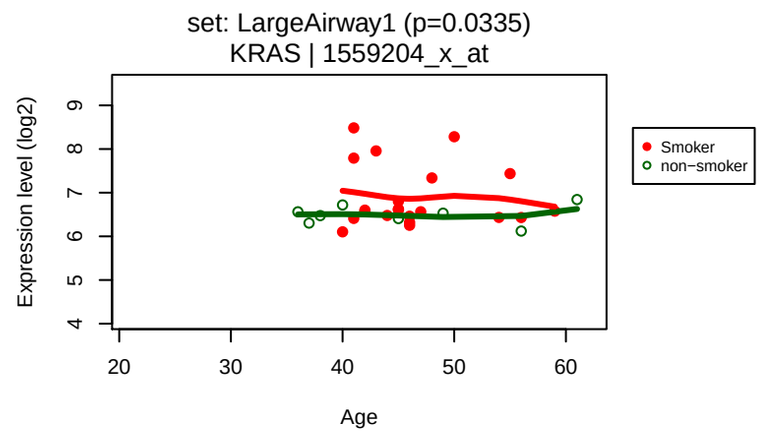

Gene expression of smokers and nonsmokers
gene: KRAS | 1559204_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1202
p=0.3904
p=0.6138
p=0.0335
p=0.3544
p=0.8424
p=0.8707
p=0.4607
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

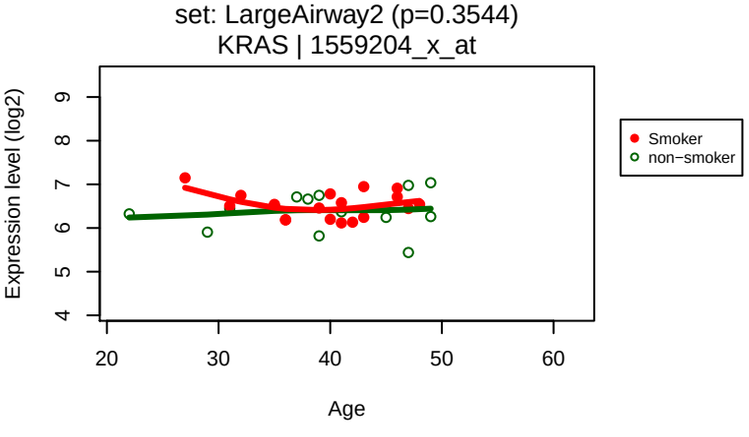

Gene expression ratio (smokers/nonsmokers)
gene: KRAS | 1559204_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1202
p=0.3904
p=0.6138
p=0.0335
p=0.3544
p=0.8424
p=0.8707
p=0.4607
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

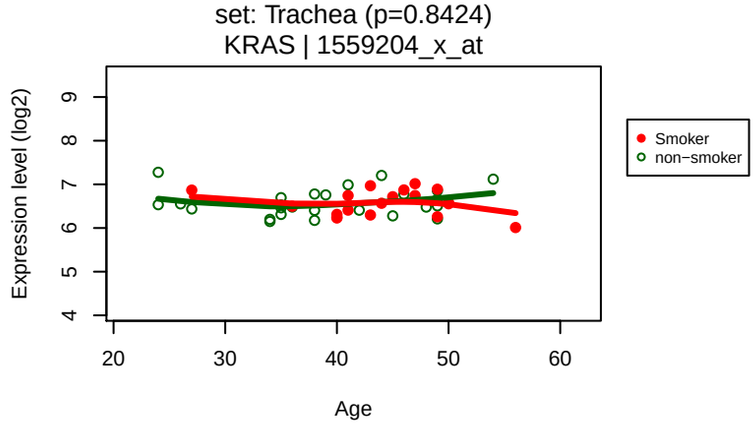

Gene expression ratio in Males
gene: KRAS | 1559204_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5882
p=0.061
p=0.8444
p=0.5615
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRAS | 1559204_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1583
p=0.0108
p=0.193
p=0.6685
SmallAirway2
LargeAirway2
Trachea
Macrophage2

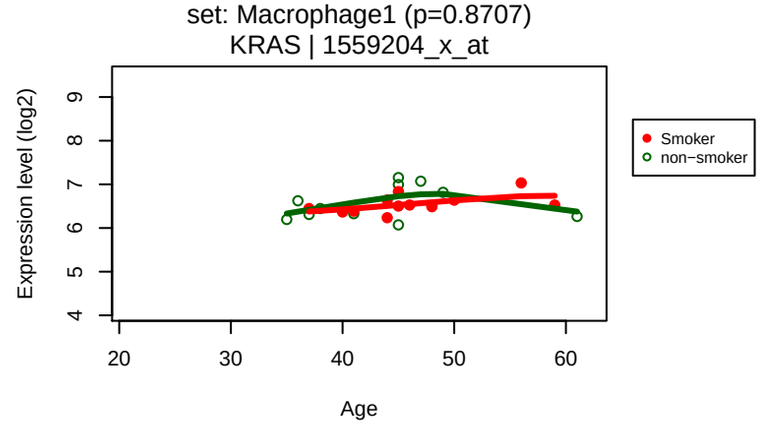

set: SmallAirway1 (p=0.1202)
KRAS | 1559204_x_at
Expression level (log2)
Age
Smoker
non−smoker

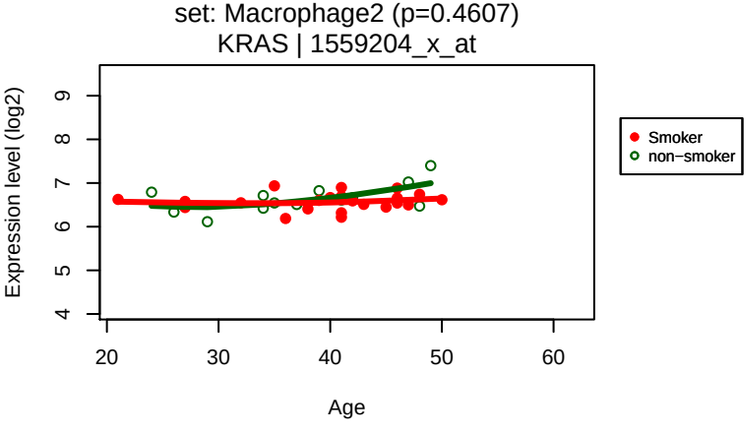

set: SmallAirway2 (p=0.3904)
KRAS | 1559204_x_at
Expression level (log2)
Age
Smoker
non−smoker

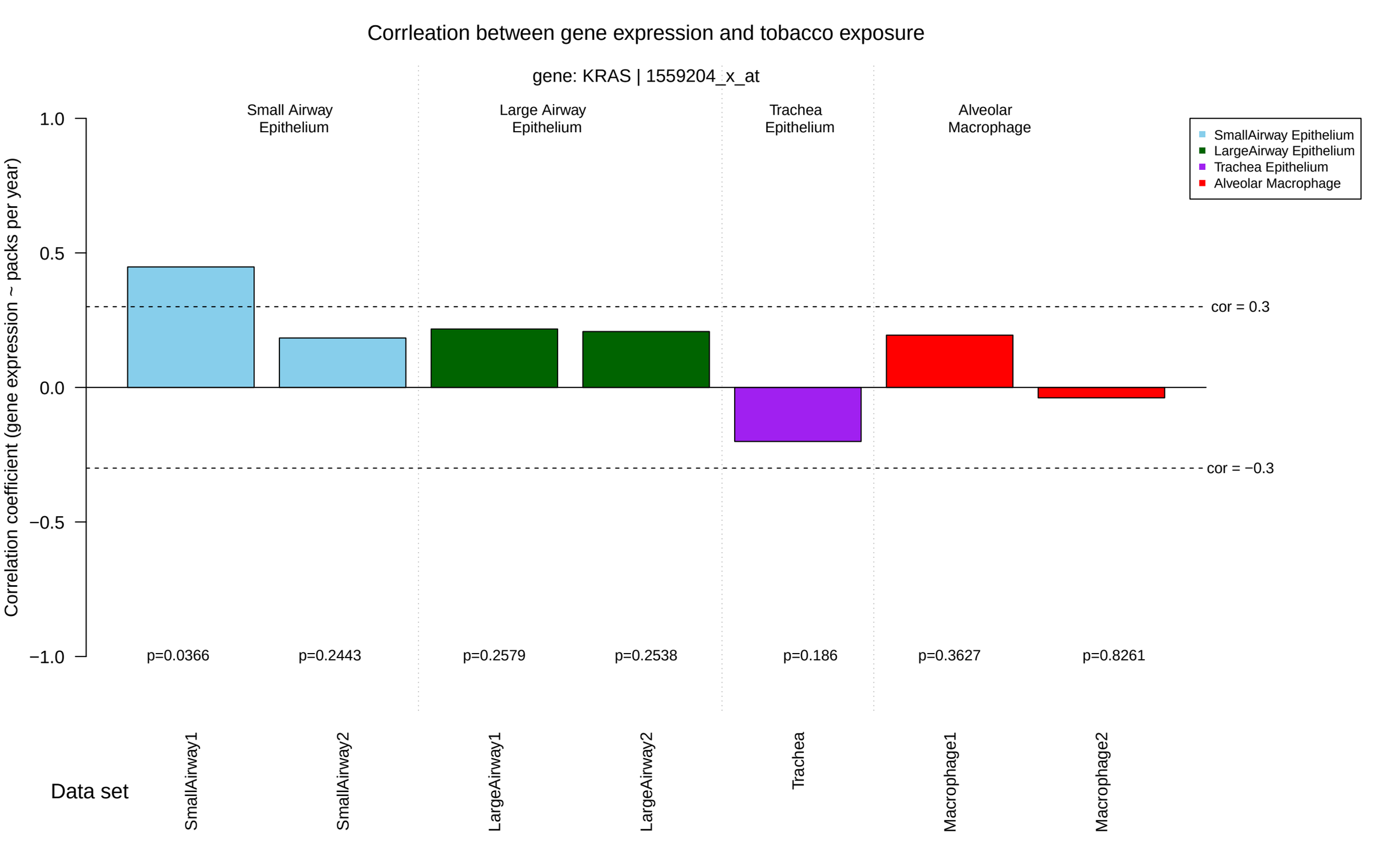

set: LargeAirway1 (p=0.0335)
KRAS | 1559204_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3544)
KRAS | 1559204_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8424)
KRAS | 1559204_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8707)
KRAS | 1559204_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4607)
KRAS | 1559204_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRAS | 1559204_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0366
p=0.2443
p=0.2579
p=0.2538
p=0.186
p=0.3627
p=0.8261
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2